TEAM NAME / NUMBER	Team 07 - TOXICITY
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PROJECT TYPE	☒ Model: train or fine-tune ☐ Evaluation ☐ System

1 · Research question

Can a Turkish BERTurk classifier fine-tuned on offensive-language labels improve macro-F1 on a manually annotated, held-out set of implicitly targeted or sarcastic attacks over a TF-IDF + logistic-regression baseline, and do token-level SHAP explanations identify human-annotated attack cues? The answer is “no” if the fine-tuned model’s macro-F1 gain is not statistically distinguishable from zero, or if SHAP highlights do not overlap annotators’ evidence more than a random-token baseline.

2 · Why it matters, and the closest work

Content moderators and researchers need to distinguish targeted mockery from harmless sarcasm and profanity-only content. Existing Turkish studies fine-tune BERT-family models for broad cyberbullying/offense classes (Bilgin & Bekar, 2025; Gökdeniz & Canbay, 2025), but do not directly establish whether a model recognizes implicit attacks or whether its explanations match human evidence. We test that narrower claim instead of assuming XAI makes predictions trustworthy.

3 · Data

Primary sources: Overfit-GM/turkish-toxic-language (Hugging Face), https://huggingface.co/datasets/Overfit-GM/turkish-toxic-language and 6-Class Labeled Turkish Tweets Dataset (Kaggle), https://www.kaggle.com/datasets/smailsariteke/trke-tweetlerin-6-snfa-gre-etiketlenmesi. The combined dataset reports 65,850 train / 13,170 validation / 8,780 test samples (87,800 total post-deduplication samples) using a 75% / 15% / 10% stratified split. Labels are multiclass toxicity target classifications across 5 distinct categories (Profanity, Insult, Racism, Sexual Harassment, and Other/Non-offensive) rather than binary offensive tags. Example from Kaggle source: “Hiç bir faydası olmayan, zararı olan, saray soytarısı Nagehan fikrini beyan ediyor” (label: Insult) and “Ulan yanan alevi yakan alevi bugün temel ile aynı çanaktan yal yiyorsun o ne olacak” (label: Racism). Annotation labels: Profanity, Insult, Racism, Sexual Content, and Harmless/Other; primary test evaluates multi-class toxic speech detection across positive and negative instances.

4 · Method and baseline

Baseline: word/character TF-IDF + logistic regression, trained only on the curated training split. Main model: BERTurk fine-tuned for binary implicit-attack vs hard-negative classification; same curated split and labels. Compute: one free Colab/Kaggle GPU or local GPU; target ≤3 GPU-hours per seed, with early stopping and a modest sequence length. Both models use identical curated examples and splits, isolating the modeling change.

5 · Evaluation

Primary metric: macro-F1 on the locked test set; also report per-class precision/recall and PR-AUC. Uncertainty: paired bootstrap 95% confidence interval for the macro-F1 difference; do not claim improvement if the interval includes zero. Explanation evaluation: token-level overlap/F1 against annotator-marked evidence, compared with a random-token baseline; manually review false positives/negatives and errors on explicit vs implicit cues. Report inter-annotator agreement before adjudication.

6 · First look

No model has been run for this project yet. This available-data check serves as an initial assessment: by combining the Overfit-GM/turkish-toxic-language (Hugging Face) and the 6-Class Labeled Turkish Tweets (Kaggle) datasets, the starting corpus comprises 87,800 post-deduplication samples. The data is partitioned into 65,850 train, 13,170 validation, and 8,780 test instances using a 75%/15%/10% stratified split. Notably, rather than broad binary offensiveness, these datasets provide multi-class toxicity classifications across five distinct categories: Profanity, Insult, Racism, Sexual Content, and Harmless/Other (e.g., distinguishing an insult like *"saray soytarısı Nagehan"* from racist statements). **Caption:** This confirms a robust, multi-class Turkish social-media starting corpus, though it does not guarantee adequate coverage of *implicit* attacks or indicate actual model performance. The curated annotation pilot and first baseline model run will be presented at the mid-project milestone.

7 · Research lead section (CSE 654 only)

Graduate research members: Ahmet Furkan Hazar, Orhun Gökdemir and Ahmet Harun Çakır. Recent work includes Bilgin & Bekar (2025), “Turkish Cyberbullying Detection with Fine-Tuned Pre-Trained Language Models” (3,388 tweets; BERTurk/DistilBERTurk/ConvBERTurk), and Gökdeniz & Canbay (2025), “Detection of Cyberbullying from Turkish Texts with Artificial Intelligence and Natural Language Processing” (TF-IDF/classical ML and FastText-based deep models). These establish Turkish-language baselines but do not evaluate explanations against human-marked evidence for implicit attacks. We will also review Mahmud, Ptaszynski & Masui (2024), “Leveraging Explainable AI and Sarcasm Features for Improved Cyberbullying Detection in Multilingual Settings” (DOI: 10.1109/DPSH60098.2024.10774626), which studies sarcasm + XAI in Bangla/Chittagonian, not Turkish. Graduate extension: compare SHAP evidence overlap to random highlights and analyze errors by explicit vs implicit cues.

8 · Biggest risk

The main risk is that available Turkish corpora tend to label generic offensiveness rather than implicit targeted attacks, and a single post may lack sufficient context to infer cyberbullying.

Mitigation Strategy: To address this, the task will be narrowly defined as detecting "implicit targeted attacks in standalone posts" rather than comprehensive cyberbullying detection, utilizing a human-annotated test set. Should annotation capacity be insufficient, the dataset size will be transparently reduced to prioritize a balanced, double-annotated test set over a larger volume of training data.

9 · Plan and roles

Verify license/schema, deduplicate, draft annotation guide, label a 50-item pilot, and run TF-IDF + logistic-regression baseline; present corpus counts and pilot agreement. By finish annotation, report baseline macro-F1, fine-tune BERTurk, and show initial error analysis. Seeded comparison, bootstrap interval, SHAP evidence-overlap analysis, graduate ablation, code and demo. Roles: graduate lead—related work, annotation protocol, experimental design/statistics; Orhun Gökdemir—data loading/curation; Burak Kurtaran—baseline and transformer fine-tuning; Ahmet Harun Çakır—XAI, evaluation and demo. All members review labels and conclusions. AI (Gemini) used for proposal drafting and research discovery; team will verify references and claims.

10 · Beyond the course (optional)

Potentially a thesis pilot or workshop submission if annotation quality is high and the final test shows a reproducible gap; novelty is not claimed until a fuller related-work review is completed.